ADITYA ANIL KOKATE

Agentic AI · Generative AI · LLM Applications · Data Science · Full-Stack Development

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Professional Summary

AI Engineer specializing in Agentic AI architectures, production multimodal pipelines, and high-throughput enterprise systems. Proven track record architecting autonomous, stateful multi-agent workflows with LangGraph, designing multi-stage vision pipelines integrating Gemini Vision, Imagen 3, and DINOv2 embeddings, and co-authoring peer-reviewed research on GraphRAG-based knowledge retrieval accepted at ICT4SD 2026. Combines research-level algorithmic rigor with hands-on production engineering, translating complex AI models into scalable FastAPI microservices with strict schema contracts, automated evaluation harnesses, and intuitive React frontends.

Technical Skills

Agentic AI & LLMs: LangGraph, Multi-Agent Workflow Design, RAG & GraphRAG Architectures, LLM API Integration (Gemini, OpenAI), Prompt Engineering & Structured Outputs, Evaluation Frameworks, Token Optimization, Transformer Mechanics
Data Science & ML: Scikit-learn, Feature Engineering, Supervised Learning, Hyperparameter Tuning (GridSearchCV), Exploratory Data Analysis, Pandas, NumPy, Statistical Modeling
Computer Vision & Multimodal: Google Gemini Vision, Vertex AI Imagen 3, Fal.ai FLUX, DINOv2 Visual Embeddings, rembg (U-2-Net), OpenCV
Full-Stack & APIs: Python FastAPI, React 18, Vite, Node.js, Express.js, REST API Architecture, SQLAlchemy, MongoDB, MySQL
Languages & Databases: Python, JavaScript / TypeScript, SQL, Java, C++, Neo4j, ChromaDB

Work Experience

Agentic & Generative AI Engineer Apr 2026 – Oct 2026 (Ongoing)
NavonLabs Pvt Ltd *Nashik, India (Hybrid)*

- Architect and deploy stateful, multi-agent systems using LangGraph and Python, orchestrating dynamic tool selection, memory retention, and fallback routing for complex domain tasks.
- Build production-grade RAG and GraphRAG pipelines, integrating automated evaluation suites to benchmark retrieval precision, context relevance, and hallucination rates across enterprise datasets.
- Lead backend development for production AI microservices using FastAPI and SQLAlchemy, standardizing structured JSON schema outputs, rate limiting, and token budget controls under live traffic.
- Design multimodal processing workflows combining vision models and vector embeddings for automated categorization, visual search, and content synthesis.

AI & Full-Stack Development Intern Dec 2024 – Jan 2025
Edunet Foundation (EY GDS & AICTE) *Remote*

- Developed and deployed full-stack MERN services in a Linux production environment, handling end-to-end build workflows, automated rollbacks, and release stability.
- Optimized MongoDB aggregation queries and established targeted database indexing, cutting query response times by 35% on high-cardinality collections.
- Engineered secure JWT authentication, role-based access control (RBAC), and RESTful API endpoints following strict security and code review standards.
- Integrated third-party payment gateways, ensuring idempotent transaction processing and API resilience under concurrent request bursts.

Key Projects

Saree Catalogue Intelligence Platform | *Confidential Client × NavonLabs* June 2026 – Present

- Engineered an end-to-end multimodal AI pipeline for automated e-commerce cataloging: background segmentation (U-2-Net), Fal.ai FLUX image inpainting, Gemini Vision for automated attribute extraction, and Imagen 3 for multi-angle product synthesis.
- Built a high-performance React 18 + Vite interface with real-time side-by-side asset comparison, backed by Python FastAPI and SQLAlchemy.
- Implemented sub-second semantic visual search across thousands of textile SKUs by indexing high-dimensional DINOv2 embeddings.
- Packaged and delivered the entire end-to-end application architecture — UI, microservices, and AI pipeline — within an aggressive investor delivery window.

eduRAG — GraphRAG Platform for Educational QA | *AISSMS IOIT, SPPU* 2025 – 2026

- Designed and implemented a multi-hop knowledge retrieval engine integrating Neo4j knowledge graphs, ChromaDB vector storage, FastAPI, and React.
- Co-authored research accepted at **ICT4SD 2026 (Goa)** detailing the underlying architecture and empirical benchmarks.
- Built automated triple extraction pipelines with confidence scoring and Leiden community clustering, improving answer fidelity and citation traceability over complex, interconnected curricula.

Sales Data Analytics & BI Dashboard | *Python, Pandas, SQL* 2024

- Built an exploratory data analysis pipeline on enterprise sales data using Python (Pandas, NumPy) and SQL, isolating seasonality, revenue concentration, and margin compression drivers.
- Developed interactive visualizations and executive summary dashboards to translate complex regression outputs into actionable commercial insights.

Travel Planner — Full-Stack Application | *MERN Stack, REST APIs* 2023

- Designed and launched a full-stack itinerary planning platform featuring dynamic routing, MongoDB schemas, and RESTful API endpoints.
- Maintained strict Git workflows, modular component separation, and clean state handling across client-server boundaries.

Education

AISSMS Institute of Information Technology, SPPU Pune, India
B.Tech in Computer Engineering (CGPA: 6.75 / 10) *2022 – 2026*

- **Core Coursework:** Artificial Intelligence, Machine Learning, Natural Language Processing, DBMS, Data Structures & Algorithms, Operating Systems, Computer Networks

G.D. Sawant College Nashik Road, India
Higher Secondary Certificate (HSC) – Science (65%) *2020 – 2022*

St. Philomena Convent High School Nashik Road, India
Secondary School Certificate (SSC) (80.80%) *2019 – 2020*

Certifications

- **AI-900: Microsoft Azure AI Fundamentals** (Azure OpenAI & Cognitive Services)
- **Building Modern Web Applications with MERN Stack** — EY GDS & AICTE
- **Joy of Computing Using Python** — NPTEL
- **Mobile App Development & UI** — MKCL